

BME680 Based Air Quality Monitoring and Temperature Alarm System Using Rugged Board and ThingsBoard

Internship Project Report

Submitted By

Aditya Shetty

Johann Jose Jinu

Project Duration

4 weeks Embedded Systems/ IOT Internship

Abstract

This project presents the design and implementation of an IoT-based environmental monitoring system using a BME680 environmental sensor interfaced with a Rugged Board running Embedded Linux. The system continuously monitors temperature, humidity, atmospheric pressure, and air quality. Sensor data is transmitted to a self-hosted ThingsBoard server through HTTP telemetry for real-time visualization and monitoring.

An SFM-27 buzzer alarm is integrated into the system to provide an immediate local alert whenever the temperature exceeds a predefined threshold of 30°C. Additionally, ThingsBoard generates remote alarms and dashboard notifications to inform users of abnormal environmental conditions.

The project demonstrates the integration of embedded Linux, sensor interfacing, network communication, cloud-based monitoring, and alarm management within a single industrial IoT application.

Objectives

The primary objectives of this project are:

1. Interface the BME680 environmental sensor with a Rugged Board using I2C communication.
2. Acquire temperature, humidity, pressure, and gas resistance measurements.
3. Estimate Air Quality Index (AQI) using BME680 gas resistance values.
4. Transmit sensor telemetry to a ThingsBoard IoT platform using HTTP.
5. Visualize sensor readings on a ThingsBoard dashboard.

6. Implement a local temperature alarm using an SFM-27 buzzer.
7. Generate remote alarms within ThingsBoard when abnormal conditions occur.
8. Gain practical experience with Embedded Linux and IoT technologies.

Hardware Components

Component	Purpose
Rugged Board	Embedded Linux Development Platform
BME680 Sensor	Environmental Monitoring
SFM-27 Buzzer	Audible Alarm
Ethernet Network	Communication with ThingsBoard
Power Supply	System Power Source

Software Components

Software	Purpose
Embedded Linux	Operating System
GCC Cross Compiler	Application Compilation
Bosch BME68x Sensor API	Sensor Driver
ThingsBoard Community Edition	IoT Dashboard and Alarm Management
HTTP Protocol	Telemetry Communication
GitHub	Source Code Management

System Architecture

The overall system architecture is shown below:

BME680 Sensor



Rugged Board



Sensor Processing Application



HTTP Telemetry



ThingsBoard Server



Dashboard Visualization

Temperature > 30°C



PD4 GPIO



SFM-27 Buzzer

Temperature > 30°C

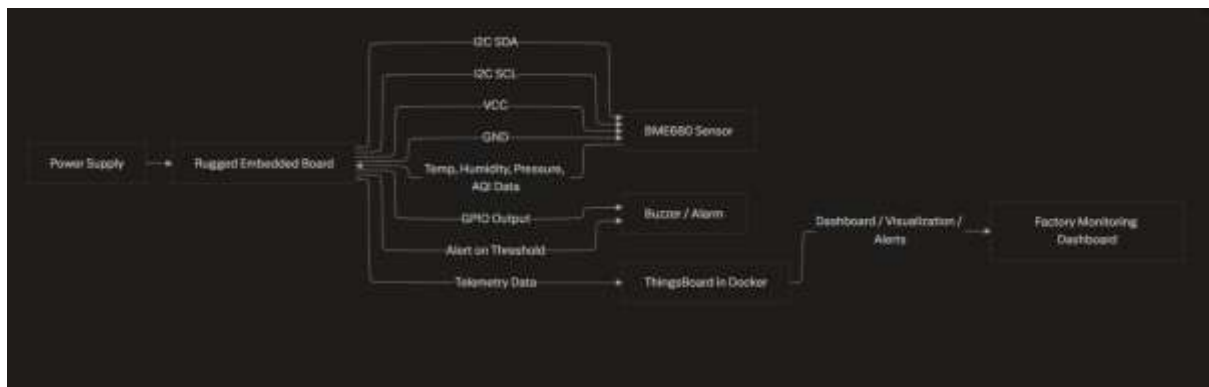


ThingsBoard Alarm Generation

Hardware connections:



Circuit Diagram:



Sensor Interfacing

The BME680 sensor communicates with the Rugged Board through the I2C interface.

Sensor Parameters Measured:

- Temperature (°C)
- Relative Humidity (%)
- Atmospheric Pressure (hPa)

- Gas Resistance (Ω)

The sensor was detected successfully using the Linux I2C tools and configured in forced mode for periodic measurements.

AQI Estimation

The BME680 does not directly measure PM2.5 or PM10 and therefore cannot calculate an official US EPA AQI value.

A custom AQI estimation algorithm was implemented using gas resistance measurements obtained from the sensor.

The AQI estimation provides a relative indication of indoor air quality and allows visualization of air quality trends on the dashboard.

AQI Categories:

- 0 – 50 : Good
 - 51 – 100 : Moderate
 - 101 – 150 : Unhealthy for Sensitive Groups
 - 151 – 200 : Unhealthy
 - 201 – 300 : Very Unhealthy
 - 301 – 500 : Hazardous
-

ThingsBoard Integration

The application establishes an HTTP connection to a self-hosted ThingsBoard server and periodically transmits telemetry data.

Telemetry Fields Sent:

- temperature
- humidity
- pressure
- aqi
- alarm

Example Telemetry:

```
{  
  "temperature": 29.5,  
  "humidity": 60.1,  
  "pressure": 1008.4,  
  "aqi": 145,  
  "alarm": 0  
}
```

The telemetry is stored by ThingsBoard and displayed on user-configurable dashboards.

Dashboard Implementation

A ThingsBoard dashboard was created to visualize environmental parameters in real time.

Dashboard Widgets:

1. Temperature Display
2. Humidity Display
3. Pressure Display
4. AQI Gauge
5. Alarm Status Widget

The dashboard provides an intuitive interface for monitoring environmental conditions.

Temperature Alarm System

A local alarm system was implemented using an SFM-27 buzzer connected to the PD4 GPIO pin.

Alarm Logic:

Temperature $\leq$ 30°C:

- Buzzer OFF

Temperature $>$ 30°C:

- Buzzer ON

This ensures immediate notification of abnormal temperature conditions.

ThingsBoard Alarm Management

ThingsBoard rule chains were configured to generate alarms automatically.

Alarm Condition:

Temperature > 30°C

Alarm Type:

High Temperature

Severity:

Critical

When the alarm condition is met:

1. The buzzer activates locally.
 2. Alarm telemetry is transmitted.
 3. ThingsBoard creates an active alarm.
 4. The dashboard displays the alarm status.
-

Results

The system successfully achieved all project objectives.

Observed Outcomes:

- Reliable communication between Rugged Board and BME680.
 - Accurate environmental data acquisition.
 - Successful HTTP telemetry transmission.
 - Real-time dashboard visualization.
 - Functional AQI estimation.
 - Immediate local buzzer alerts.
 - Automated alarm generation in ThingsBoard.
 - Stable operation over extended periods.
-

Skills Acquired

During this project, the following skills were developed:

- Embedded Linux Development
 - Cross Compilation
 - I2C Communication
 - Sensor Interfacing
 - Embedded C Programming
 - HTTP Networking
 - ThingsBoard IoT Platform
 - Alarm Management Systems
 - Git and GitHub
 - Industrial IoT Application Development
-

Future Improvements

1. Integration of Bosch BSEC library for advanced IAQ calculations.
 2. Addition of PM2.5 and PM10 sensors for true AQI measurement.
 3. Email and SMS alert notifications.
 4. MQTT-based telemetry communication.
 5. Data logging and analytics.
-

Conclusion

An IoT-based environmental monitoring and alarm system was successfully developed using a BME680 sensor, Rugged Board, and ThingsBoard platform. The system provides real-time environmental monitoring, air quality estimation, local audible alerts, and cloud-based alarm management. The project demonstrates practical implementation of Embedded Linux, sensor interfacing, networking, and Industrial IoT technologies, providing valuable experience in the design and deployment of smart monitoring systems.

References

1. Bosch BME680 Sensor Datasheet
2. Bosch BME68x Sensor API Documentation

3. [Embedded Linux Documentation](#)
4. [ThingsBoard Community Edition Documentation](#)
5. [Linux I2C Subsystem Documentation](#)
6. [GCC Cross Compilation Documentation](#)